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In [12]: # Import required libraries
import numpy as np
import pandas as pd
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load Breast Cancer dataset
data = load_breast_cancer()

# Create DataFrame
df = pd.DataFrame(data.data, columns=data.feature_names)

# Add target column
df['target'] = data.target

print("First 5 Rows of Dataset:")
print(df.head())

# Features and target
X = data.data
y = data.target

# Split dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create SVM model with Gaussian (RBF) Kernel
svm_model = SVC(kernel='rbf')

# Train the model
svm_model.fit(X_train, y_train)

# Predict on test data
y_pred = svm_model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("\nAccuracy:", accuracy)

# Confusion Matrix
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Classification Report
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

First 5 Rows of Dataset:

	mean radius	mean texture	mean perimeter	mean area	mean smoothness \
0	17.99	10.38	122.80	1001.0	0.11840
1	20.57	17.77	132.90	1326.0	0.08474
2	19.69	21.25	130.00	1203.0	0.10960
3	11.42	20.38	77.58	386.1	0.14250
4	20.29	14.34	135.10	1297.0	0.10030

	mean compactness	mean concavity	mean concave points	mean symmetry \
0	0.27760	0.3001	0.14710	0.2419
1	0.07864	0.0869	0.07017	0.1812
2	0.15990	0.1974	0.12790	0.2069
3	0.28390	0.2414	0.10520	0.2597
4	0.13280	0.1980	0.10430	0.1809

	mean fractal dimension	...	worst texture	worst perimeter	worst area \
0	0.07871	...	17.33	184.60	2019.0
1	0.05667	...	23.41	158.80	1956.0
2	0.05999	...	25.53	152.50	1709.0
3	0.09744	...	26.50	98.87	567.7
4	0.05883	...	16.67	152.20	1575.0

	worst smoothness	worst compactness	worst concavity	worst concave points \
0	0.1622	0.6656	0.7119	0.2654
1	0.1238	0.1866	0.2416	0.1860
2	0.1444	0.4245	0.4504	0.2430
3	0.2098	0.8663	0.6869	0.2575
4	0.1374	0.2050	0.4000	0.1625

	worst symmetry	worst fractal dimension	target
0	0.4601	0.11890	0
1	0.2750	0.08902	0
2	0.3613	0.08758	0
3	0.6638	0.17300	0
4	0.2364	0.07678	0

[5 rows x 31 columns]

Accuracy: 0.9473684210526315

Confusion Matrix:

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[[37  6]
 [ 0 71]]
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Classification Report:

	precision	recall	f1-score	support
0	1.00	0.86	0.93	43
1	0.92	1.00	0.96	71
accuracy			0.95	114
macro avg	0.96	0.93	0.94	114
weighted avg	0.95	0.95	0.95	114

In []: